

Cowen-Set Membership Does Not Determine the Blaschke–Potapov Degree

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Status. Candidate full counterexample, likely valid, pending expert review. Problem 4.2 of Abhinand–Curto–Prasad has a negative answer, already for analytic diagonal monomial symbols. For this family we also determine the exact set of determinant degrees occurring in the matrix Cowen set.

1 Source question and answer

For $\Phi \in L^\infty(\mathbb{T}, M_n)$, the source defines

$$\mathcal{E}(\Phi) = \{K \in H^\infty(\mathbb{T}, M_n) : \|K\|_\infty \leq 1 \text{ and } \Phi - K\Phi^* \in H^\infty(\mathbb{T}, M_n)\}.$$

Problem 4.2 asks whether, for a normal symbol Φ , the presence of a finite Blaschke–Potapov product $B \in \mathcal{E}(\Phi)$ forces

$$\text{rank}[T_\Phi^*, T_\Phi] = \deg(\det B).$$

Problem 4.2. *Let $\Phi \in L^\infty(\mathbb{T}, M_n)$ be a normal symbol and suppose that $\mathcal{E}(\Phi)$ contains a finite Blaschke–Potapov product B . Does it follow that*

$$\text{Rank}[T_\Phi^*, T_\Phi] = \deg(\det B)?$$

Figure 1: Problem 4.2 on page 11 of arXiv:2605.02214.

The answer is no. Indeed, for any $m, n \geq 1$ and $k > m$, take

$$\Phi_m(z) = z^m I_n, \quad B_k(z) = z^k I_n.$$

Then $B_k \in \mathcal{E}(\Phi_m)$, but the two quantities in the proposed identity are mn and kn . Thus one may take the concrete block example $\Phi(z) = zI_2$, $B(z) = z^2I_2$, giving ranks/degrees 2 and 4.

Proof intuition

The Cowen-set condition is stable under adding extra inner factors on the analytic side. For a monomial symbol $z^m I_n$, every $z^k I_n$ with $k \geq m$ still makes $\Phi - B\Phi^*$ analytic. The Toeplitz commutator, however, only sees the fixed symbol and has rank mn . The proposed equality therefore cannot hold for an arbitrary member B of the Cowen set. More sharply, for diagonal monomials, analyticity forces a minimal inner divisor; all additional inner degree is optional.

2 A full counterexample

Let S denote the unilateral shift on $H^2(\mathbb{T})$. We use the convention $[A^*, A] = A^*A - AA^*$.

Theorem 1 (Monomial counterexample). *For every $m, n \geq 1$ and every $k > m$, the pointwise normal symbol $\Phi_m(z) = z^m I_n$ and the finite Blaschke–Potapov product $B_k(z) = z^k I_n$ satisfy*

$$B_k \in \mathcal{E}(\Phi_m), \quad \text{rank}[T_{\Phi_m}^*, T_{\Phi_m}] = mn, \quad \deg(\det B_k) = kn.$$

Consequently Problem 4.2 is false, including when $n \geq 2$.

Proof. On $\mathbb{T}$ we have $\Phi_m^*(z) = z^{-m} I_n$, and hence

$$\Phi_m - B_k \Phi_m^* = (z^m - z^{k-m}) I_n \in H^\infty(\mathbb{T}, M_n).$$

The matrix B_k is inner, so $\|B_k\|_\infty = 1$; this proves $B_k \in \mathcal{E}(\Phi_m)$. Pointwise, Φ_m is a scalar multiple of the identity and is therefore normal (in fact unitary on $\mathbb{T}$).

Under $H^2(\mathbb{T}, \mathbb{C}^n) \cong H^2(\mathbb{T}) \otimes \mathbb{C}^n$,

$$T_{\Phi_m} = S^m \otimes I_n.$$

Since $S^{*m} S^m = I$ and $I - S^m S^{*m}$ is the projection onto $\text{span}\{1, z, \dots, z^{m-1}\}$,

$$[T_{\Phi_m}^*, T_{\Phi_m}] = (I - S^m S^{*m}) \otimes I_n$$

has rank mn . Finally $\det B_k = z^{kn}$, whose degree is kn . Because $k > m$, the proposed equality fails. $\square$

3 Exact degree spectrum for diagonal monomials

The same mechanism yields a sharp result, not merely one counterexample.

Theorem 2 (Minimum-degree theorem). *Let*

$$D(z) = \text{diag}(z^{m_1}, \dots, z^{m_n}), \quad m_j \geq 1, \quad r = \sum_{j=1}^n m_j.$$

Then

$$\text{rank}[T_D^*, T_D] = r$$

and the finite Blaschke–Potapov products in $\mathcal{E}(D)$ have exactly the determinant degrees

$$\{\deg(\det B) : B \text{ finite Blaschke–Potapov}, B \in \mathcal{E}(D)\} = \{r, r+1, r+2, \dots\}.$$

In particular, the commutator rank is the minimum, not the common value, of the determinant degrees in the Cowen set.

Proof. The Toeplitz operator is the orthogonal direct sum $T_D = \bigoplus_{j=1}^n S^{m_j}$. The computation in Theorem 1 gives

$$[T_D^*, T_D] = \bigoplus_{j=1}^n (I - S^{m_j} S^{*m_j}),$$

so its rank is r .

Now let a finite Blaschke–Potapov product $B = (B_{ij})$ belong to $\mathcal{E}(D)$. The (i, j) entry of BD^* on $\mathbb{T}$ is $B_{ij}(z)z^{-m_j}$. Since $D - BD^*$ has no negative Fourier coefficients, neither can this entry: the analytic entry D_{ij} has none to cancel. Thus every Taylor coefficient of B_{ij} below order m_j vanishes. Equivalently, the j th column of B is divisible by z^{m_j} , and

$$B = CD$$

for an analytic rational matrix function C .

Both B and D are unitary almost everywhere on $\mathbb{T}$, so $C = BD^*$ is unitary there. Hence C is rational inner. Taking determinants gives

$$\det B = (\det C)z^r.$$

The scalar rational inner function $\det C$ has a nonnegative integer degree; therefore every such B satisfies $\deg(\det B) \geq r$.

Conversely, for each $d \geq 0$ set

$$B_d(z) = \text{diag}(z^{m_1+d}, z^{m_2}, \dots, z^{m_n}).$$

This is a finite Blaschke–Potapov product and

$$B_d D^* = \text{diag}(z^d, 1, \dots, 1),$$

so $D - B_d D^*$ is analytic and $B_d \in \mathcal{E}(D)$. Moreover $\deg(\det B_d) = r + d$. This realizes every integer at least r . $\square$

4 Quantifier audit and sharp repair

Problem 4.2 names a particular product B already known to lie in $\mathcal{E}(\Phi)$ and asks whether its degree must equal the rank. Theorems 1 and 2 refute exactly that literal statement. There is no ambiguity from block size: the examples work for all n , and in particular for every $n \geq 2$.

The paper’s central conjecture has a different, existential quantifier: under hyponormality and finite-rank hypotheses, it asks whether there *exists* a finite Blaschke–Potapov product in $\mathcal{E}(\Phi)$ whose determinant degree is the commutator rank. Our examples do not refute that conjecture. For the diagonal symbols above, $B = D$ belongs to $\mathcal{E}(D)$ and has degree r , so it is exactly the desired witness.

Thus the sharp repaired statement for this natural class is

$$\text{rank}[T_D^*, T_D] = \min\{\deg(\det B) : B \text{ finite Blaschke–Potapov in } \mathcal{E}(D)\},$$

whereas equality for every available B is false. Whether this minimum-degree formulation holds for broader normal symbols is not claimed.

5 Verification, scope, and novelty

The proof uses only Fourier coefficients and the standard Hardy basis. The review-critical checks are: the sign of the self-commutator, the right-column exponents in BD^* , and the universal-versus-existential quantifier. All three computations are displayed explicitly.

This packet completely answers Problem 4.2 negatively and gives an exact classification for diagonal monomial symbols. It does not settle the paper’s central existential conjecture for arbitrary symbols.

Local run indexes were searched by arXiv identifier, title, and the core Cowen-set/degree terminology. Bounded primary-source searches through August 11, 2026 found the source paper but no later answer or correction. The argument is elementary and may be folklore; novelty is asserted only as an explicit answer to this 2026 problem and remains subject to expert review.

References

- [1] M. Abhinand, R. E. Curto, and T. Prasad, *Hyponormal block Toeplitz operators with finite rank self-commutators*, arXiv:2605.02214 (2026), Problem 4.2, p. 11.